# Project Findings & Methods Log: Robust and Explainable Classical ML for Tomato Leaf Disease

Oct 13, 2026 · @Manahil Kamran

## 1. Project overview

This project measures how well lightweight classical models (hand-crafted color, texture and shape features feeding a Random Forest) cope with real-field variation in tomato leaf photos, and which features stay trustworthy when conditions change. This log records what we did, what we found, what went wrong and what limits the conclusions, so each point can be lifted into the mid-semester report and the final paper.

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**Research questions (from the synopsis).**

1. How well do simple classical models cope with the visual variation found in real tomato fields?
2. Which image features can still be trusted when lighting, contrast or background change?

**Objectives (from the synopsis).**

1. Build a baseline on color, texture and shape features with Random Forest as the main model, and SVM and Logistic Regression only as a brief check.
2. Measure the performance loss under illumination/contrast and background perturbations.
3. Run a feature-group ablation (color, texture, shape, combined) to see which cues stay reliable.
4. Test how stable feature importance is across folds and perturbed inputs.

**Deliverables and how they are graded** (from the term-project brief).

| Deliverable | Due | Marks | What is graded |
|---|---|---|---|
| Synopsis (2 pages) | Sep 25, 2026 | 10 | Submitted. Motivation, objectives, dataset exploration, metrics |

| Deliverable | Due | Marks | What is graded |
|---|---|---|---|
| Mid-semester report | Nov 13, 2026, 23:59 PST | 10 | Literature survey, methodology tied to course concepts, preliminary results, challenges handled honestly |
| Final term paper | Dec 24, 2026, 23:59 PST | 20 | IEEE format, 5-6 pages. Abstract, novelty, results and error analysis, limitations |

The brief asks the mid-semester report to include revisions to the original plan with reasoning. Section 3 records the main one: the dataset turned out to be smaller than the synopsis stated.

## 2. Dataset

We use the Tomato Leaf Disease Classification Dataset in Pakistan (Malik, Ali and Ali, Mendeley Data v2, DOI 10.17632/3mbnb82mxd.2, February 2026), specifically its `Dataset (raw)` folder, which holds 830 original photos. The 7,200 images quoted in the dataset description and in our synopsis belong to the augmented version.

**What the dataset documentation states.**

| Property | Value |
|---|---|
| Collection | Real agricultural fields in Pakistan, natural daylight, uncontrolled backgrounds |
| Capture device | Smartphone camera |
| Labeling | Manual, based on visible symptoms |
| Format and resolution | JPG, stated as uniform resolution for all images |
| Classes (6) | Early Blight, Late Blight, Septoria Leaf Spot, Leaf Mold, Yellow Leaf Curl Virus, Healthy |
| Stated total | 7,200 images |
| License | CC BY 4.0 |

**What the folders actually contain** (from the Drive folder and the audit).

| Folder | Content | Used? |
|---|---|---|
| `Dataset (raw)` | One sub-folder per class, no train/val/test split, random file names. 830 images found | Yes, main data |
| `Augmented Dataset` | `train`, `val`, `test`, each with one folder per class. Files named like `100_46341_aug_5aug62.jpg` and `100_46341aug1.jpg`, i.e. derived copies of a source photo | No (see section 3) |
| `Dataset_Description.pdf`, `License.txt`, `README.md` | Documentation | Reference |

**Class folder names** (raw folder, as spelled by the authors) and the label we assign.

| Raw folder | Label used in code |
|---|---|
| `Tomato_Early_blight_leaf` | early_blight |
| `Tomato_leaf_late_blight` | late_blight |
| `Tomato_Septoria_leaf_spot` | septoria_leaf_spot |
| `Tomato_mold_leaf` | leaf_mold |
| `Tomato_leaf_yellow_curl_virus` | yellow_leaf_curl_virus |
| `Tomato_Healthy_leaf` | healthy |

The augmented folders spell two names differently (`Tomato_septora_leaf_spot`, `Tomato_Early_Blight`), so the class matcher accepts both spellings.

**Why this dataset fits the question.** The variability we want to study (changing light, busy backgrounds, partly visible symptoms) is already in the photos, so perturbations are layered on top of realistic images instead of on clean lab shots. No published benchmark exists for it yet (checked at synopsis time), so our numbers serve as a first baseline, not a replication.

## 3. Findings so far

The dataset holds 830 original photos, not 7,200, so the synopsis assumptions about size, balance and split no longer hold and the plan has been revised. The numbers below come from our first real audit run (`00_audit_dataset.py --hash`) on `Dataset (raw)` on Oct

13, 2026.

| Audit measure | Value |
|---|---|
| Images found in `Dataset (raw)` | 830 |
| Classes recognised | 6 |
| Ignored files or unmatched folders | None reported |
| Pre-made train/val/test folders | None |
| Files marked as augmented (by name) | 0.0% |
| Near-duplicate pairs found | 55 |
| Of which pair two different classes | 9 |
| Independent photos after merging copies | 785 (from 830) |
| Largest group of copies | 3 images |

**Findings.**

1. **Real size.** 830 originals versus 7,200 stated. 7,200 / 830 is about 8.7 augmented images per original, which is consistent with the augmented folder being the 7,200-image set. We have not yet audited the augmented folder to confirm its count.
2. **Synopsis assumptions that no longer hold.** The synopsis states 1,200 images per class, a fixed 840/240/120 split per class, balanced classes (so no class weighting) and a 120-image test set per class. These describe the augmented set. For the raw set the per-class counts are 119 (early blight), 137 (healthy), 113 (late blight), 186 (leaf mold), 195 (septoria) and 80 (yellow leaf curl), an imbalance of 2.4x; the split is our own, and the test set has 121 images (12 to 29 per class). This is a plan revision to report, with this reasoning.
3. **Copies inside the raw set.** 45 of 830 images (5.4%) are near-copies of another image (resized, recompressed, flipped or slightly re-lit). We merge them into one group so they cannot sit on both sides of the split.
4. **Possible label noise.** 9 of the 55 near-duplicate pairs carry two different class labels. Either the same photo is filed under two diseases or the matcher paired two similar-looking photos. Update after viewing six of them in results/duplicate_examples.png: they are the same photo filed under two diseases, so the matcher is not at fault. File names agree: a file named EB_(969).jpg (EB = early blight) sits in the septoria folder, and IMG_20251102_155805.jpg exists in both the early blight and the late blight folders. The true label of these photos cannot be known.

5. **Augmented set risk (not yet quantified on real data).** Augmented files share a stem with their source photo ( `100_46341` appears in the raw folder and as `100_46341aug1.jpg` ). If the official train/val/test split was made after augmenting, siblings of one photo can sit in train and test and inflate scores. Its test images also already contain brightness, flip and rotation changes, which confounds a perturbation study. We chose raw to avoid both.
6. **Provenance questions from file names.** In the one raw folder we looked at (Early Blight, 8 visible files) some names look like phone camera-roll files ( `_storage_emulated_0_DCIM_Camera_IMG_20251102_...` ), while others look like a different camera or web source ( `20080526-_DSC4211.jpg` , `1576389-early%20blight%20symptoms.jpg` ). The documentation says smartphone field photos at uniform resolution. Image sizes now confirm the concern: widths run from 150 to 4,416 px, so the resolution is not uniform, and the median size differs sharply by class (septoria 0.33 megapixels, late blight 0.48, leaf mold 0.81, yellow leaf curl 0.94, early blight 6.75, healthy 9.53). Some septoria images are stock photos (an agency watermark is visible) and the leaf mold images carry camera timestamp overlays. EXIF data has not been checked. See problem 14 in section 11.
7. **Likely same-plant bursts.** Camera-roll names with close timestamps ( `IMG_20251102_155855` , `_161045` , `_161217` ) suggest several shots of one plant in one session. These are different photos, so the visual grouping will not merge them. They can still make test images easier than a truly new field would be. See limitations.

**After the first full run (Oct 13).** Stages 00 to 11 have now run twice on the real data (v1 on Oct 13, v2 on Oct 14). Results are in section 13, problems and their solutions in section 11, and the plan from here in sections 15 and 16. The most important finding is that image size and file size alone predict the class (macro-F1 0.53, against 0.61 to 0.66 for the full model), so a source shortcut is the main threat to any claim about disease recognition.

## 4. Data preparation

We re-split the 830 photos ourselves into 70% train, 15% validation and 15% test, per class, keeping near-copies of one photo together, so that test scores are not inflated by seeing the same photo in training.

**Steps (all in `src/data.py` , `src/duplicates.py` , run by `scripts/01_prepare_data.py` ).**

1. **Scan and label.** Walk `Dataset (raw)` , map each folder name to a class by keyword, record path, label and a source group.
2. **Hash every image.** Resize to 32x32 grey, take a 2-D DCT, keep the low-frequency 8x8 block without the DC term (63 values), and set each bit by comparison with the median. This is a perceptual hash (pHash): JPEG noise and brightness scaling barely

move it. Hashes are also computed on the horizontally, vertically and doubly flipped image, so flipped copies match.

3. **Candidate pairs.** Two images are candidates if their hashes differ in at most 10 of 63 bits.
4. **Verify pairs.** A candidate is accepted only if the 32x32 colour thumbnails (after the same flip) have normalised correlation of at least 0.98. This removes false matches between different leaves that merely look alike.
5. **Group.** Accepted pairs of the same class are merged into one group (union-find). Pairs across classes are counted as possible label noise and not merged.
6. **Split by group.** Within each class, groups are shuffled with seed 42 and filled greedily into train (70%), validation (15%) and test (15%) by image count. No group is ever shared between splits.
7. **Cache.** Every image is resized once to 256x256 (area interpolation, aspect ratio not preserved) and stored in one array in manifest order. All later stages read this cache.
8. **Commit the manifest.** `data/manifest.csv` (path, label, split, class id, group, is_aug) is committed once so all three of us use the identical split.

**Parameters.**

| Parameter | Value | Where |
|---|---|---|
| Split ratio | 70 / 15 / 15 per class | `dataset.group_split` |
| Seed | 42 | `seed` |
| pHash block, bits | 8x8, 63 bits | `duplicates.hash_size` |
| Hamming threshold | 10 bits | `duplicates.hamming_threshold` |
| Correlation threshold | 0.98 | `duplicates.min_correlation` |
| Cached image size | 256x256 | `preprocess.image_size` |

**Safeguards.** The code prints a warning if any group exceeds 2% of the data (a sign of chained false matches) and if a class has too few groups for all three splits. It writes `results/grouping_summary.json` with the pair count, groups before and after, and the largest group.

**How the method was validated (synthetic test only).** We built a fake dataset with the raw folder layout and 72 planted copies (flipped, recompressed at lower quality, slightly brighter). The final method recovered 72 of 72 and kept all 72 pairs in the same split. These images are synthetic, so the thresholds are not tuned on real photos; they need checking against `results/duplicate_examples.png` on the real data.

**Version 2 changes (run on Oct 14).** The first full run led to five changes, all switchable in config.yaml. (1) Photos that appear under two diseases are dropped together with all their copies (duplicates.drop_cross_class), and the list is written to results/dropped_cross_class.csv. (2) Class weights are set to balanced for all three models. (3) Stage 10 trains a classifier on image width, height, aspect ratio and file size alone, to measure how much of the class the image source reveals. (4) Stage 11 repeats the grouped split with five seeds and reports mean and standard deviation. (5) The leaf mask is compared across three settings: the HSV gate, the gate refined by GrabCut, and no mask. Outcome: 17 images (9 conflicting pairs plus all their copies) were removed, leaving 813 images (570 train, 125 validation, 118 test), and the test set has no copy of any training or validation photo.

## 5. Preprocessing and leaf segmentation

The dataset has no leaf masks, so we segment the leaf with a simple unsupervised colour rule; its quality limits both the shape features and how realistic the background perturbation is.

**Method (`src/segment.py`).**

1. Convert the 256x256 image to HSV (OpenCV scale: hue 0-179).
2. Keep pixels with saturation at least 40, value at least 40 and hue between 20 and 100 (yellow to green). Brown soil and brown lesions fall outside the hue range.
3. Morphological opening with a 5x5 ellipse and closing with a 15x15 ellipse.
4. Keep connected components at least as large as the larger of 20% of the biggest component and 1% of the image.
5. Fill holes, so lesions fully enclosed by healthy tissue return to the mask.
6. If the mask covers under 5% of the image, treat segmentation as failed and use the whole image.

A slower GrabCut refinement is available (`method: grabcut`) so the two can be compared.

**Where the mask is used.** All colour and texture statistics are computed on leaf pixels only; shape and lesion features use the mask outline; the background perturbation replaces everything outside it. Segmentation is re-run on every perturbed image, so robustness numbers measure the whole pipeline and not only the classifier.

**Known failure modes.**

- Green backgrounds (grass, neighbouring plants, other leaves) pass the hue gate, so the mask can include them or the rule falls back to the whole image.
- Brown lesions touching the leaf edge are excluded from the mask, so their area is lost.

- Heavy shadow or glare can push value or saturation below the thresholds.
- A thin halo of original background can remain after background replacement.

**What to report once the real run is done.** `results/eda_segmentation_stats.json` gives the mean mask fraction and the share of images where segmentation fell back to the whole image; `results/eda_segmentation.png` shows 12 examples. Report the fallback rate in the paper, and look at the images before trusting it.

On our synthetic test images the first version labelled almost every image as all leaf (49% fell back); adding the hue gate fixed it. That says nothing about real photos.

**Real-data result (first run, Oct 13).** Across the 830 images the mean mask covers 83% of the frame (median 89%). 30.6% of masks cover more than 95% of the image and 6.4% fell back to the whole image. In results/eda_segmentation.png the outlines often follow soil, shadow or neighbouring leaves rather than the target leaf, and dark leaves in shade are cut out. In dense-canopy photos "the leaf" is not well defined, so any mask is a judgement. The v2 run compared it with GrabCut and with no mask (section 13): no mask is as accurate on clean images, and GrabCut is the most robust to lighting changes. GrabCut starts from the hue-gate mask, so on photos where that mask already covers most of the frame it has little to refine; the no-mask run tests whether segmentation helps at all.

## 6. Feature engineering

Each image becomes a vector of 105 hand-crafted features in three groups: 47 colour, 38 texture and 20 shape/edge/lesion, all computed on the leaf mask ( `src/features.py` ).

| Group | Count | Features | Why it should help |
|---|---|---|---|
| Colour | 47 | Mean, standard deviation and skewness of each channel in RGB, HSV and LAB (27); 12-bin hue histogram; 8-bin saturation histogram | Chlorosis, necrosis and browning change hue and saturation; LAB separates lightness from colour |
| Texture | 38 | Uniform LBP at two scales, (P=8, R=1) giving 10 bins and (P=16, R=2) giving 18 bins; GLCM contrast, dissimilarity, homogeneity, energy and correlation at distances 1 and 3, averaged over 4 angles (10), on 32 grey levels | Mould, spotting and blight patterns differ in surface texture |

| Group | Count | Features | Why it should help |
|---|---|---|---|
| Shape, edge, lesion | 20 | Leaf area fraction, circularity, solidity, extent, eccentricity (5); 7 log-scaled Hu moments; Canny edge density, gradient mean and standard deviation, Laplacian variance (4); non-green lesion fraction, log lesion count, largest and mean lesion fraction (4) | Curl virus deforms leaves; spots differ in size and count |

**Lesion features.** Inside the eroded leaf mask, a pixel counts as non-green when its excess-green index (2G - R - B over R + G + B) is below 0.15. Small regions (under 20 pixels) are dropped before counting.

**Implementation notes.**

- Features are extracted in parallel with joblib and cached as one matrix per split in manifest order, so they are computed once.
- Feature names start with the group (`color_`, `texture_`, `shape_`), so ablations are column selections.
- Non-finite values (rare) are replaced with 0 and reported.
- This stage is CPU-bound; a GPU does not help.

**Course concepts to cite.** Feature engineering and curse of dimensionality (105 features on roughly 580 training images), correlated features (RGB/HSV/LAB statistics overlap), and noise sensitivity of derived features.

## 7. Models and tuning protocol

Random Forest is the main model; an RBF SVM and a Logistic Regression act as a brief cross-check. Hyper-parameters are chosen on the validation split only, and the test split is used once, for final numbers (`src/models.py`, `scripts/04_baseline.py`).

| Model | Settings | Searched grid | Course concept |
|---|---|---|---|
| Random Forest | 300 trees, seed 42 | max depth {none, 20, 40}, max features {sqrt, 0.3}, min leaf size {1, 2} | Bagging, variance reduction, bias-variance trade-off |
| SVM | RBF kernel, standardised inputs | C {1, 10, 100}, gamma {scale, 0.01, 0.001} | Margin maximisation, kernels |

| Model | Settings | Searched grid | Course concept |
|---|---|---|---|
| Logistic Regression | L2 penalty, multinomial, standardised inputs | C {0.01, 0.1, 1, 10} | Regularisation, linear decision boundary |

**Protocol.**

1. Fit each configuration on the train split and score macro-F1 on the validation split; keep the best.
2. Refit the chosen configuration on train plus validation.
3. Evaluate once on the clean test split and report accuracy, macro-F1 with a 95% bootstrap interval, per-class precision, recall and F1, and the confusion matrix.
4. Report test metrics again on the subset of test images that have no near-duplicate or shared group in train or validation, when any are flagged.
5. Plot a Random Forest learning curve (train and validation macro-F1 against training-set size) to show where the bias-variance trade-off sits.

**Why Random Forest as the main model.** Hand-crafted features are heterogeneous, differently scaled and correlated; trees need no scaling, handle mixed feature types, give impurity importances and are cheap enough for low-end devices, which is the deployment setting in the synopsis.

**Class balance.** The synopsis argued class weighting was unnecessary because every class had 1,200 images. That argument applied to the augmented set. The raw set is imbalanced by 2.4x (80 yellow leaf curl images against 195 septoria), so the v2 run adds class weights (balanced) and keeps macro-F1 as the headline metric.

## 8. Experiments: ablation and robustness

Two experiments reuse the same trained models: a feature-group ablation on clean data, and a robustness test that perturbs only the test images (`scripts/05_ablation.py`, `scripts/06_robustness.py`).

**Ablation.** Seven feature sets (colour, texture, shape, the three pairs, and all) crossed with the three models. Each model is refit on train plus validation using the hyper-parameters tuned on all features, and scored once on the clean test set. Reusing the all-feature hyper-parameters is a simplification; it slightly favours the full set.

**Perturbations.** 15 conditions, applied to test images only (`src/perturb.py`). Each is deterministic for a given seed, image and level.

| Kind | Levels | Implementation | What it imitates |
|---|---|---|---|
| Brightness | x0.5, x0.7, x1.3, x1.6 | Multiply pixel values, clip to 0-255 | Dim light, strong sun |
| Contrast | x0.5, x0.7, x1.5, x2.0 | Scale deviation from the image mean | Haze, flat or harsh light |
| Shadow | strength 0.3, 0.5, 0.7 | Soft diagonal darkening with random direction and edge position | A shadow across the leaf |
| Background | grey, noise, soil, grass | Replace pixels outside the leaf mask with a synthetic texture, edge blurred | Different ground or plants behind the leaf |

**Robustness protocol.**

1. Train every model once on clean train plus validation.
2. For each condition, perturb the test images, re-run segmentation, re-extract all features, and predict.
3. Report accuracy and macro-F1 per condition and the drop from the clean score (clean minus perturbed), per model and feature set. The Random Forest is evaluated on every feature set; SVM and Logistic Regression on all features.
4. Error analysis for the worst condition: confusion matrix, per-class F1 drop and the shift in predicted-class shares.

**Why test-only perturbation.** The question is how a model trained on the data as collected copes with new conditions, so training stays clean. Training on perturbed images is an optional follow-up, not part of the current design.

**Caveat.** Brightness and contrast are global; real fields mix lighting effects. Background replacement is only as good as the mask (section 5), and a failed mask means that image's background is left unchanged. On the real data the clean mask covers 86% of the frame on average, so only about 14% of an image can be replaced. For the grass background the mean leaf-area fraction after replacement is 1.0: the painted grass is green, so the hue gate absorbs it into the leaf mask. That result describes the segmentation as much as the classifier.

# 9. Explainability stability protocol

We treat explainability as something to measure: how much the feature ranking of the Random Forest changes between data folds and between clean and perturbed inputs (`src/explain.py`, `scripts/07_importance_stability.py`).

**Importance measures.**

- **Impurity importance** from the forest itself.
- **Permutation importance**: shuffle one feature (or one whole feature group) across the evaluation rows and record the drop in the mean probability the model gives to the true class.
- **Grouped permutation**: colour, texture and shape are each shuffled jointly. Colour statistics in RGB, HSV and LAB overlap, so single-feature rankings split credit arbitrarily among correlated features and are expected to be unstable; the group view asks a steadier question.

**Two stability tests.**

| Test | Setup | Compared |
|---|---|---|
| Across folds | 5-fold stratified cross-validation on train plus validation, with folds that keep each photo group together; one forest per fold | Rankings from every pair of folds |
| Across perturbations | The fixed forest from the baseline, importance recomputed on the clean test set and on each of the 15 perturbed test sets | Each perturbed ranking against the clean ranking |

**Agreement statistics.** Spearman and Kendall rank correlation over all 105 features, and Jaccard overlap of the top-10 and top-20 features. For the three groups, the same statistics are reported over the group ranking.

**Why the true-class probability, not F1.** On a small test set most features cause no change in the predicted label when shuffled, so an F1 drop is exactly zero for many features and rank correlations become undefined or meaningless. The mean true-class probability changes smoothly, so the ranking has few ties. (This was found while testing on synthetic data, where F1-based importances produced undefined correlations.)

**Interpretation caution.** High rank agreement shows the model uses the same cues, not that those cues are medically meaningful. Low agreement under a perturbation shows the model's reasoning changed, which is the finding we want to report.

# 10. Evaluation metrics and statistical notes

Macro-F1 is the headline metric because every class counts equally; accuracy is reported only for comparison with earlier tomato-disease papers.

| Metric | Definition or use | Why |
|---|---|---|
| Accuracy | Share of test images classified correctly | Comparability with prior work |
| Macro-F1 | Mean of per-class F1 | Easy classes cannot hide weak ones; missing a disease is costly |
| Per-class precision, recall, F1 | From the test predictions | Shows which diseases fail |
| Confusion matrix | Row-normalised counts | Shows which diseases are mixed up and under which condition |
| Robustness drop | Clean score minus perturbed score, for accuracy and macro-F1 | Direct measure of robustness |
| Importance stability | Spearman, Kendall, top-k Jaccard | Explainability as a number |

**Uncertainty.** Test macro-F1 carries a 95% bootstrap interval (500 resamples of the test images). With a test set of 121 images (12 to 29 per class) the intervals are wide (random forest macro-F1 0.61, interval 0.51 to 0.69): differences of a point or two between models or feature sets are not evidence of a real difference. The bootstrap resamples images and ignores the small group structure, which is acceptable because most groups contain a single image.

**Not yet done.** No significance test between models (a paired bootstrap or McNemar test on the same test images would be the natural choice), and a single split only. Stage 11 (new) repeats the grouped split with five seeds to show how much results depend on which images landed in the test set; it has not been run yet.

**Leak-free reporting.** Test metrics are also computed on the images with no near-duplicate or shared group in train or validation, and both numbers are reported when any image is flagged.

# 11. Problems encountered and how each was handled

Twenty-seven problems so far. Problems 1 to 5 and 13 to 27 come from the real dataset (the audit, v1 on Oct 13 and v2 on Oct 14); the rest were found while testing the code on synthetic images and are listed because they shaped design decisions.

| # | Problem | Seen on | Why it matters | Resolution |
|---|---|---|---|---|
| 1 | Synopsis says 7,200 images, 1,200 per class, fixed 840/240/120 split. The raw folder has 830 images | Real data | Test set is 121 images; classes range from 80 to 195 images; synopsis claims need revising | Use the 830 originals, make our own grouped split, report the revision with reasoning |
| 2 | Augmented set: files are derived copies of source photos (shared file-name stem) | Real data (folder names and file names) | Siblings of one photo may sit in train and test, inflating scores; test images already carry brightness, flip and rotation changes | Do not use it for evaluation; use raw. Auditing it ( `split_strategy: official` ) remains possible |
| 3 | Raw set has no train/val/test split | Real data | We must create one without leakage | Grouped 70/15/15 split per class, seed 42, manifest committed |
| 4 | Random file names, so copies cannot be grouped by name | Real data | 45 of 830 images (5.4%) are near-copies; 9 pairs cross classes | Perceptual-hash candidates verified by thumbnail correlation (section 4); cross-class pairs flagged for review |
| 5 | Folder spelling differs between folders ( `septora` vs `Septoria` , `Early_Blight` vs `Early_blight_leaf` ) | Real data | A class would be silently skipped | Class matcher accepts both spellings; the audit lists unmatched folders |

| # | Problem | Seen on | Why it matters | Resolution |
|---|---|---|---|---|
| 6 | First near-duplicate hash (difference hash) scored identical images 28-41 bits apart after recompression | Synthetic | Copies would be missed | Switched to a DCT perceptual hash |
| 7 | The coarse hash also matched different leaves, risking chains that merge many images into one group | Synthetic | A giant group would land in one split and unbalance it | Added a thumbnail-correlation check (at least 0.98) and a warning when one group exceeds 2% of the data |
| 8 | First segmentation labelled saturated brown soil as leaf (49% whole-image fallback) | Synthetic | Features and background replacement would be meaningless | Added a hue gate (20-100) and a larger closing kernel |
| 9 | F1-based permutation importance gave many exact zeros, so rank correlations were undefined | Synthetic | Stability statistic would be meaningless | Importance measured as drop in true-class probability |
| 10 | Ordinary cross-validation would put copies of a photo in different folds | Design review | Optimistic scores and inflated importance stability | Folds keep each photo group together |
| 11 | Reading thousands of files straight from Google Drive is slow | Colab setup | Wasted GPU-free session time | Unzip the dataset to the notebook's local disk first |
| 12 | Colab deletes local files when a session ends | Colab setup | Lost cache, features and models | Notebook backs `work/` up to Drive and restores it |

| # | Problem | Seen on | Why it matters | Resolution |
|---|---|---|---|---|
| 13 | The same photo appears under two diseases (9 near-duplicate pairs with two labels; file names agree, for example EB_(969).jpg in the septoria folder) | Real data (first run) | Label noise caps accuracy; 6 pairs straddle splits, and 3 test and 2 validation images have a conflicting copy | Done in v2: 17 images (9 pairs plus their copies) removed and listed in dropped_cross_class.csv; no test image now has a copy in train or validation (3 before) |
| 14 | Image size is not uniform (150 to 4,416 px wide) and differs by class (median 0.33 MP septoria, 9.53 MP healthy); stock photos and timestamped camera images are mixed in | Real data (first run) | The class may be predictable from the image source, and resizing to 256x256 destroys lesion texture in large photos | Stage 10 confirmed the shortcut: metadata alone gives macro-F1 0.533 (chance 0.167). Next: resolution-controlled variants, source-aware splits and shortcut-pruned features (section 15) |
| 15 | Leaf masks are unreliable: 30.6% cover more than 95% of the image, 6.4% fall back to the whole image, and outlines often follow soil or shadow | Real data (first run) | Shape and lesion features and the background perturbation inherit the errors | Compared in v2: clean accuracy is the same for all three and GrabCut is the most robust to lighting (section 13). Next: measure mask quality on about 40 hand-drawn masks |
| 16 | The background perturbation reaches little of the image (the clean mask covers 86% of the frame) and painted grass is absorbed into the mask (leaf-area fraction 1.0) | Real data (first run) | Background robustness is partly a statement about segmentation, not only about the classifier | The no-mask run could not show it: the perturbation uses the same mask setting, so it replaces nothing. Next: one fixed reference mask for all perturbations (section 15) |

| # | Problem | Seen on | Why it matters | Resolution |
|---|---|---|---|---|
| 17 | Random forest overfits: train+val accuracy 0.99 against test 0.62; learning curve training F1 1.00 against validation 0.58 | Real data (first run) | Model differences are small and the 95% intervals overlap (RF, SVM and LR all score 0.59 to 0.61 macro-F1) | Partly done: five repeated splits show model differences are within noise, so no model ranking is claimed. Next: regularised forest, feature selection, patch-based samples, train-only augmentation |
| 18 | Class imbalance of 2.4x and small test classes (12 yellow leaf curl images) | Real data (first run) | Per-class F1 swings by several points on a single image | Balanced class weights are on in v2; their separate effect was not measured (no unweighted v2 run). Next: run both over repeated splits |
| 19 | Robustness numbers mix classifier and mask: contrast 0.5 and shadow 0.7 lower the mean leaf area from 0.86 to 0.76 and 0.75 | Real data (first run) | A drop under perturbation may come from the mask, not the classifier | Compared in v2: leaf area under shadow 0.7 falls from 0.83 to 0.71 (HSV) and from 0.80 to 0.62 (GrabCut), yet GrabCut scores higher, so mask area alone does not explain the drop. Next: reference-mask perturbation |
| 20 | Group-level stability is trivially 1.0 because there are only 3 feature groups | Real data (first run) | The grouped stability number carries no information | v2 gave 1.0 again. Report per-feature statistics, say why the grouped one is uninformative, and use 10 to 15 sub-groups and correlation clusters (section 15) |

| # | Problem | Seen on | Why it matters | Resolution |
|---|---|---|---|---|
| 21 | Single-split numbers overstate differences: seed 42 is the best of five random-forest splits, and model and feature-set gaps (0.01 to 0.03) are smaller than the split-to-split std (0.03 to 0.05) | Real data (v2) | A rise or fall of a few points between runs or methods is not evidence | Report mean and std over at least 10 repeated grouped splits, with paired tests on matched splits (section 15) |
| 22 | A v1 robustness claim did not replicate: logistic regression lost 0.167 at contrast 2.0 in v1 and 0.435 in v2 (random forest 0.379 and 0.391) | Real data (v1 against v2) | Model rankings under perturbation from one split are unreliable | Extract perturbed features for all images once, then score robustness on every repeated split |
| 23 | The perturbation uses the same mask setting as the features, so with no mask the background conditions change nothing (RF scores 0.640, the clean value, under all four) | Real data (v2) | Background robustness cannot be compared across masks | Perturb with one fixed reference mask (GrabCut, or hand-drawn masks on an evaluation subset) whatever mask the features use |
| 24 | Shape features add nothing: shape alone scores 0.35 to 0.43, and colour+texture is 0.008 to 0.017 above all features | Real data (v2) | Extra dimensions raise variance and instability without gain | Make shape an optional ablation; test lesion statistics separately; select features inside each training fold |
| 25 | The top-10 important features change between folds (Jaccard 0.55 for impurity, 0.78 for permutation) | Real data (v2) | A list of top features is not a stable explanation | Importance at the level of correlated feature clusters, SHAP with bootstrap intervals, and stability measures beyond top-k (section 15) |

| # | Problem | Seen on | Why it matters | Resolution |
|---|---|---|---|---|
| 26 | The repeated splits reuse hyper-parameters tuned on the seed-42 validation split | Design review (v2) | Slight optimism, and the other seeds' test images overlap the images used for tuning | Nested tuning: choose parameters inside each training split |
| 27 | The dataset page says every image is a handheld-phone field photo from Pakistan with no online sources, but the data contain a stock-photo watermark, sizes from 150 to 4,416 px and web-style file names | Real data | The source and licence of some images are unclear, and an external dataset may overlap with ours | Hash-match against any external dataset before using it; ask the dataset authors; state it in the limitations |

**Still open.** Closed in v2: 13. Partly addressed: 15, 17, 18, 19 and 21. Open: 14, 16, 20 and 22 to 27. Problem 14 is the main threat to the research claim and comes first in section 15.

## 12. Limitations and threats to validity

The main limits are a small dataset, an imperfect leaf mask, and grouping that catches copies but not repeat shots of the same plant. Each is stated so the paper's conclusions are not stretched beyond them.

### Data

- **Small sample.** About 830 photos (785 independent) in six classes. The test set has 121 images (12 to 29 per class), so every score has a wide interval (random forest macro-F1 0.61, 95% interval 0.51 to 0.69).
- **Class imbalance.** Raw counts range from 80 to 195 per class (2.4x). The v2 run uses balanced class weights, but the smallest classes still have only 12 test images.
- **Single split.** One grouped split with seed 42. Five repeated splits (v2) give a standard deviation of 0.03 to 0.05 in macro-F1, so single-split differences below that are noise.
- **Same-plant leakage not removed.** Grouping catches resized, recompressed, flipped and re-lit copies. It does not catch a different photo of the same leaf or plant (for example burst shots with close timestamps), so test images may be easier than a truly

new field. Grouping by capture time, where file names carry it, is a possible refinement.

- **Possible label noise.** 9 near-duplicate pairs carry two labels, so the same photo was filed under two diseases. These photos are removed in v2, but a mislabelled photo with no duplicate would go unnoticed.
- **Provenance.** Some file names suggest sources other than a phone in the authors' fields. Partly confirmed: image sizes run from 150 to 4,416 px and differ by class, and some images are stock photos or carry camera timestamps. The dataset page states that every image is a handheld-phone field photo from Pakistan with no online sources, which the watermark, file names and sizes contradict. EXIF data is unchecked and the source of each image is unknown; the dataset authors could be asked.
- **Resizing.** Images are squashed to 256x256 without keeping aspect ratio, identically in every split. Fine texture is lost at this size.

### Method

- **Segmentation is unsupervised and approximate.** It works against soil but not against green backgrounds, and it drops brown lesions on the leaf edge. Shape and lesion features and the background perturbation inherit these errors. On the real data 30.6% of masks cover more than 95% of the frame and 6.4% fall back to the whole image.
- **Synthetic perturbations.** Brightness, contrast, shadow and background replacement are global or single-pattern; real fields combine effects. Background replacement leaves a thin halo and does nothing where segmentation failed.
- **Test-only perturbation.** We do not train on perturbed images, so the results describe a model trained on data as collected. They do not show the best achievable robustness.
- **Hyper-parameters tuned on all features** are reused in the ablation and robustness tests, which slightly favours the full feature set.
- **Hash thresholds** were set on synthetic images. On the real data 52 of the 55 pairs have Hamming distance 0 and the largest group has 3 images, so the matches look like exact copies; near-copies at larger distances would still be missed.
- **Correlated features.** Colour statistics overlap, so single-feature importance is unstable by construction; the grouped view is the safer reading.
- **No significance tests** between models or conditions yet.

### Added after the first full run

- **Source shortcut.** Image size differs by class (median 0.32 megapixels for septoria against 9.53 for healthy), and image size and file size alone reach a macro-F1 of 0.53. No claim about disease recognition is safe until the control experiments in section 15 are done.

- **Segmentation and robustness are entangled.** Perturbations change the leaf mask (mean leaf area under shadow 0.7: 0.71 for HSV, 0.62 for GrabCut, against 0.83 and 0.80 clean), so a drop in score is not purely the classifier's.
- **The perturbation mask is not independent.** The perturbation uses the same mask setting as the features, so background conditions cannot be compared across masks (problem 23).
- **Mask comparison is one split.** GrabCut's robustness advantage rests on one test set of 118 images; it is a lead, not a result.
- **Shared tuning.** The repeated splits reuse hyper-parameters tuned on the seed-42 validation split, which slightly favours that seed.
- **Fixes not separated.** v2 removed 17 images and turned on class weights at the same time, so the effect of each alone is unknown.

**Scope**

- One crop, one country, one dataset. Conclusions do not transfer to other crops or regions without testing. The PlantCity out-of-distribution test mentioned in the synopsis is optional and not yet done.
- Classical features and models only, by design. A CNN reference would show what is traded away, but is not part of the current plan.

## 13. Results (v1 on Oct 13, v2 on Oct 14)

Two full runs on the real data: v1 (Oct 13, 830 images) and v2 (Oct 14, 813 images, after removing 17 images with conflicting labels and turning on balanced class weights). Every number is copied from the listed result files. Single-split numbers (seed 42) are noisy: five repeated splits show a standard deviation of 0.03 to 0.05 in macro-F1, so differences smaller than that are not evidence.

| Model | v1, seed 42 (830 images) | v2, seed 42 (813 images) | v2, mean ± std over 5 splits | v2 train+val accuracy |
|---|---|---|---|---|
| Random forest | 0.610 (0.507 to 0.691) | 0.660 (0.560 to 0.736) | 0.606 ± 0.036 | 1.000 |
| SVM (RBF) | 0.606 (0.505 to 0.691) | 0.656 (0.558 to 0.743) | 0.616 ± 0.046 | 0.999 |
| Logistic regression | 0.592 (0.501 to 0.675) | 0.633 (0.541 to 0.717) | 0.587 ± 0.044 | 0.827 |

Test macro-F1, with the 95% bootstrap interval in brackets. Seed 42 is the best of the five splits for the random forest (0.660 against a mean of 0.606), so the v1 to v2 rise of 0.05 cannot be credited to the fixes: the two runs differ in test images, class weights and removed images, and the step is about 1.4 standard deviations of split-to-split noise. v2 has no test image with a copy in train or validation (3 in v1). Five-fold grouped cross-validation gives a random forest macro-F1 of 0.575 ± 0.029 (0.565 ± 0.030 in v1). In the v2 learning curve the training F1 stays at 1.00 while validation F1 plateaus between 0.54 and 0.59 from 225 training images on (0.42 at 54 images), so more data of this kind helps little. The three models are statistically indistinguishable: the gaps between them (0.02 to 0.03) are smaller than the standard deviation over splits (0.04 to 0.05).

**Feature sets (v2, mean ± std of test macro-F1 over 5 splits).**

| Feature set | RF | SVM | LR |
|---|---|---|---|
| Colour (47) | 0.521 ± 0.028 | 0.539 ± 0.035 | 0.482 ± 0.042 |
| Texture (38) | 0.488 ± 0.029 | 0.512 ± 0.054 | 0.546 ± 0.045 |
| Shape (20) | 0.426 ± 0.036 | 0.348 ± 0.042 | 0.401 ± 0.026 |
| Colour + texture (85) | 0.621 ± 0.028 | 0.633 ± 0.052 | 0.595 ± 0.053 |
| All (105) | 0.606 ± 0.036 | 0.616 ± 0.046 | 0.587 ± 0.044 |

Colour plus texture is as good as or better than all features for every model (by 0.008 to 0.017, inside the noise), so shape adds nothing. Each group alone is far below the combination, so colour and texture carry complementary information. In v1 (one split) the full set was best by 0.02 to 0.03 and every interval overlapped, so the five-split result is the more reliable one.

**Leaf mask comparison (v2, seed 42, test macro-F1).** "Perturbed" is the mean over the 11 brightness, contrast and shadow conditions; the four background conditions are left out because the no-mask run cannot be perturbed that way (see below).

| Mask | RF clean | SVM clean | LR clean | RF perturbed | SVM perturbed | LR perturbed |
|---|---|---|---|---|---|---|
| HSV gate | 0.660 | 0.656 | 0.633 | 0.503 | 0.400 | 0.472 |
| GrabCut after HSV | 0.638 | 0.721 | 0.649 | 0.525 | 0.505 | 0.514 |
| No mask | 0.640 | 0.647 | 0.610 | 0.455 | 0.387 | 0.454 |

On clean images the three masks are statistically the same (95% intervals of about ±0.09 overlap), and no mask is as accurate as the HSV gate: segmentation does not clearly help accuracy. Under lighting changes GrabCut has the highest mean score for all three

models, and the SVM gap is large (brightness 0.5: 0.471 with GrabCut, 0.217 with HSV, 0.150 with no mask; shadow 0.7: 0.516, 0.292, 0.236). This is one split and needs repeated splits before it is a claim. The mean leaf area also differs (clean 0.83 for HSV, 0.80 for GrabCut), and the grass background is absorbed into the mask for both (leaf area 1.0). In the no-mask run the background conditions score exactly the clean value (0.640 for RF) because the perturbation uses the same mask setting and so replaces nothing: that is a design flaw, not robustness (problem 23).

**Source shortcut check (v2, stage 10).** A random forest that sees only image width, height, aspect ratio, pixel count, file size and bytes per pixel reaches test macro-F1 0.533 (accuracy 0.559) and grouped cross-validation macro-F1 0.482. The majority-class accuracy is 0.237 and chance macro-F1 is 0.167. The metadata alone therefore reproduces roughly 80 to 85% of the full model's score (0.660 on the same test split; 0.575 in cross-validation). Median image size differs by class:

| Class | Median megapixels | Size range (10th to 90th percentile) |
|---|---|---|
| Healthy | 9.53 | 0.05 to 12.53 |
| Early blight | 9.00 | 0.09 to 12.53 |
| Yellow leaf curl | 0.94 | 0.13 to 12.53 |
| Leaf mold | 0.81 | 0.68 to 2.77 |
| Late blight | 0.41 | 0.05 to 10.23 |
| Septoria | 0.32 | 0.05 to 12.00 |

The two best classes for the random forest (healthy F1 0.82, leaf mold 0.79) are the ones with the most distinctive source signature: healthy is mostly large phone-sized photos, and leaf mold sits in a narrow size range with a median aspect ratio of 0.75 (the other classes are 1.0 or more) and timestamp overlays. This is consistent with a shortcut but does not prove one: the hand-crafted features could be reading the disease as well. It does mean no claim about disease recognition is safe until the control experiments in section 15 are done.

**Robustness (v2, seed 42, HSV mask, macro-F1 drop from clean).** Random forest: contrast 2.0 0.391, shadow 0.7 0.262, contrast 0.5 0.217, contrast 1.5 0.187, brightness 1.6 0.161, and for backgrounds 0.009 (gray), 0.088 (soil), 0.127 (grass) and 0.140 (noise). The SVM and logistic regression suffer as much: SVM brightness 0.5 0.439 and contrast 2.0 0.413, LR contrast 2.0 0.435. Lighting changes, not backgrounds, break the models. Under contrast 2.0 the predictions collapse towards leaf mold (from 25% to 54% of all predictions); late blight F1 falls from 0.60 to 0.00 and healthy from 0.82 to 0.08. The v1 run showed the same pattern (leaf mold 25% to 64%). One v1 statement did not replicate: v1 reported that

logistic regression lost far less than the random forest at contrast 2.0 (0.167 against 0.379); in v2 it lost 0.435 against 0.391. Model rankings under perturbation are therefore not reliable from one split (problem 22).

**Importance and its stability (v2).** Over five grouped folds the rank correlation of permutation importance is 0.889 ± 0.012 (v1 0.864) and of impurity importance 0.894 ± 0.013 (v1 0.905), so the overall ranking is stable. The identity of the top features is not: top-10 overlap (Jaccard) is 0.78 for permutation and 0.55 for impurity (v1 0.73 and 0.74), and top-20 overlap 0.67 and 0.80. The grouped stability of 1.0 is trivial because there are only three groups (problem 20). The top features by permutation drop are colour LAB B standard deviation (0.042), texture GLCM correlation at distance 3 (0.031), HSV saturation standard deviation (0.022) and GLCM correlation at distance 1 (0.019). Permuting a whole group costs colour 0.17 to 0.20, texture 0.08 to 0.10 and shape 0.03 to 0.05. GLCM correlation depends on image sharpness and therefore on native resolution, which makes it a plausible channel for the source shortcut (a hypothesis to test, section 15). Under perturbation the rank correlation with the clean ranking falls to 0.20 (contrast 0.5) and 0.36 (contrast 2.0) but stays between 0.75 and 0.96 for shadows and backgrounds.

**Error analysis (v2).** Most confused: early blight predicted as septoria (6 of 15 test images), yellow leaf curl as septoria (4 of 12), septoria as late blight (4 of 28), and leaf mold and septoria in both directions (3 each). Per-class random forest F1: healthy 0.82, leaf mold 0.79, early blight 0.62, late blight 0.60, yellow leaf curl 0.58, septoria 0.56.

| Experiment | Source files in results/ | Status |
|---|---|---|
| Dataset audit and grouping | dataset_audit.json, grouping_summary.json, dropped_cross_class.csv | v1 and v2 done |
| EDA and image sizes | eda_samples.png, eda_class_stats.csv, eda_brightness_by_class.png, eda_image_sizes.csv | v1 and v2 done |
| Segmentation check | eda_segmentation_stats.json, eda_segmentation.png | v1 and v2 done (30.6% and 30.1% of masks cover more than 95% of the frame) |
| Near-duplicates and leakage | duplicates_summary.json, duplicate_examples.png, test_leakage_flags.csv | v1 and v2 done |

| Experiment | Source files in results/ | Status |
|---|---|---|
| Baseline | baseline_test_metrics.csv, baseline_per_class_test.csv, confusion_*test.png, baseline_tuning*.csv, learning_curve_rf.csv | v1 and v2 done |
| Ablation | ablation.csv, fig_ablation.png | v1 and v2 done (single split) |
| Robustness | robustness.csv, robustness_drop.csv, fig_robustness_curves.png, fig_robustness_drop_heatmap.png | v1 and v2 done (single split) |
| Importance stability | importance_stability_cv.json, importance_stability_perturbed.csv, importance_top_features.csv, fig_group_importance.png, fig_importance_stability.png | v1 and v2 done |
| Error analysis | error_pairs.csv, misclassified_examples.png, confusion_rf_worst_condition.png, per_class_f1_drop_worst.csv | v1 and v2 done |
| Source shortcut check | source_shortcut_result.json, source_shortcut_by_class.csv, source_shortcut_metadata.csv | v2 done |
| Repeated splits | repeated_splits.csv, repeated_splits_summary.csv | v2 done (5 seeds, parameters from seed 42) |
| Mask comparison | results_seg_grabcut/, results_seg_none/ (baseline and robustness only) | v2 done (single split) |

**Questions each result should answer in the write-up.**

1. Does the clean score drop when the 125-image test set replaces a leaky or augmented one? Compare the all-test and leak-free scores.
2. Which perturbation family hurts most, and which classes lose most?
3. Does a single feature group carry the model, and does that group collapse under a particular perturbation?
4. Is the importance ranking stable across folds, and does it survive perturbations? Report both the per-feature and the grouped view.

5. Which diseases are confused with which, and does the confusion change under perturbation?

## 14. Reproducibility, paper mapping and open items

Every experiment is reproducible from one repository, one config file and one committed split; this section also maps the log onto the report and paper sections and lists what remains.

**Reproducibility.**

- **Code:** `github.com/manahilcpkhan/tomato-robust-ml`. Logic lives in `src/`, one runnable script per stage in `scripts/` (00 audit through 11 repeated splits), a thin Colab runner in `notebooks/`.
- **One config:** `config.yaml` holds the seed (42), paths, feature settings, search grids, perturbation levels and thresholds. Personal paths go in a git-ignored override.
- **One split:** `data/manifest.csv`, committed once.
- **Large files** (image cache, features, models) stay out of Git and are backed up to Drive; small results are committed.
- **Software:** Python with NumPy, pandas, SciPy, scikit-learn, scikit-image, OpenCV, joblib and matplotlib (`requirements.txt`). CPU only; no GPU is used.
- **Work split:** Member A data and perturbation, Member B features and models, Member C evaluation, explainability and report integration.

**Where each part goes.**

| This log | Mid-semester report (Nov 13) | Final paper (Dec 24) |
|---|---|---|
| 1, 2 | Introduction | Introduction, dataset |
| 3, 11 | Challenges and plan revisions | Dataset section, limitations |
| 4, 5, 6, 7 | Methodology (tie each choice to a course concept) | Method |
| 8, 9, 10 | Methodology, experiment design | Experiments |
| 13 | Preliminary results | Results and error analysis |
| 12 | Challenges | Limitations |

**Not covered here: the literature survey.** Both the synopsis and the mid-semester report are graded on it. The synopsis cites four works (PlantVillage, PlantDoc, the Pakistan tomato dataset, PlantCity). A fuller survey should cover lab versus field datasets,

handcrafted-feature and classical-ML plant disease work, robustness and distribution shift in plant disease recognition, and feature-importance stability. We have not collected or checked any further references.

**Open items.**

- [x] ~~Stages 00 to 11 run twice (v1 on Oct 13, v2 on Oct 14); problems 13 to 27 logged with their outcomes.~~
- [x] ~~Inspect the 9 cross-class pairs, check image sizes, check the segmentation fallback rate, decide on class weights.~~
- [x] ~~Run stage 10: metadata alone reaches macro-F1 0.53, so the source shortcut is confirmed.~~
- [ ] Phase 1 (to Oct 24): repeated nested protocol, source-shortcut controls (resolution-matched variants, source clusters, shortcut-pruned features), extended perturbation set with a fixed reference mask and perturbed features for all images (section 15).
- [ ] Phase 2 (to Nov 8): mask quality on about 40 hand-drawn masks, illumination handling, feature selection, regularised models, boosting, one CNN-embedding reference, train-only augmentation.
- [ ] Mid-semester report (Nov 13): write up with the Phase 1 and 2 results and the literature survey.
- [ ] Phase 3 (Nov 14 to Dec 10): patch-based samples, explainability study, external test set (hash-match first), significance tests.
- [ ] Audit Augmented Dataset to quantify its leakage and confirm its 7,200 count; map augmented files to their raw sources before any use as training copies.
- [ ] Check EXIF data and ask the dataset authors about image sources.
- [ ] Write the literature survey.

## 15. Recommended pipeline from here

Three results shape the plan. (1) Image size and file size alone reach macro-F1 0.53, so a source shortcut may explain much of the 0.61 to 0.66 score. (2) Split-to-split noise (std 0.03 to 0.05) hides most model and feature differences. (3) Lighting changes, not backgrounds, break the models, and the mask matters less than expected. The pipeline first removes the shortcut and the noise, then compares masks, illumination handling, features, models and augmentation under one protocol, and puts the explainability study on top.

| # | Stage | What to do | Evidence it answers | Priority |
|---|---|---|---|---|
| 1 | Evaluation protocol | Repeated grouped splits (10 seeds) or 5x3 repeated grouped cross-validation; tune inside each training split; report mean and std; paired tests on matched splits with Holm correction; print the metadata-only baseline as a reference row in every table | Problems 21 and 26: seed 42 flatters the forest and the gaps sit inside the noise | Phase 1, must |
| 2 | Source-shortcut controls | (a) Resolution-matched variants: V0 as now; V1 every image degraded to the same effective resolution before features (downscale to 128 px, then back to 256); V2 only images with both sides at least 500 px. (b) Source clusters from size, aspect ratio, compression and file-name pattern, then leave-one-cluster-out evaluation and accuracy per cluster. (c) Correlate every feature with log pixel count and bytes per pixel, drop those above 0.5 in absolute Spearman correlation (GLCM correlation is the first suspect) and re-score. (d) Re-encode all images identically (same resize, same JPEG quality) before extracting features. If scores fall towards the metadata baseline, the shortcut explained them | Problem 14: metadata alone 0.533; GLCM correlation is a top feature | Phase 1, must |

| # | Stage | What to do | Evidence it answers | Priority |
|---|---|---|---|---|
| 3 | Label and data quality | Inspect the 30 to 50 most confidently wrong test predictions by eye; optionally apply confident learning (for example cleanlab) to cross-validated probabilities; log every removal with its reason | Problem 13: the 9 conflicting pairs were found by chance | Phase 1, should |
| 4 | Perturbation framework | Add gamma, white-balance colour cast, Gaussian and motion blur, JPEG quality, additive noise, downscale-then-upscale (tests the resolution shortcut directly), partial occlusion, rotation and scale; 3 to 5 severities; a few combined pairs; one fixed reference mask for background replacement; extract perturbed features for all images once so every split can be scored | Problems 16, 22 and 23 | Phase 1, must |
| 5 | Leaf masks | Compare none, HSV, GrabCut after HSV, and a pretrained segmenter run once and cached (for example SAM or a salient-object network). Score every mask against about 40 hand-drawn leaf masks (IoU) so the choice is justified. Keep a separate lesion mask (non-green pixels inside the leaf) for lesion features | Problems 15 and 19: GrabCut most robust to lighting, no mask as accurate on clean images | Phase 2, should |

| # | Stage | What to do | Evidence it answers | Priority |
|---|---|---|---|---|
| 6 | Illumination handling | Before features: gray-world or shades-of-gray colour constancy, CLAHE on lightness, per-image intensity normalisation. In features: chromaticity, hue and saturation histograms, LAB a and b without L. Evaluate as test-time fixes with no retraining | Largest drops: contrast 2.0, shadow 0.7, brightness 0.5 | Phase 2, must |
| 7 | Features | Keep colour and texture as the core, shape optional. Add multi-scale LBP and Gabor or wavelet energies at a fixed scale, and lesion statistics (count, area fraction, size spread, lesion colour against healthy tissue). Then cut 105 to about 30 to 40 features by correlation clustering plus importance, done inside each training fold | Problem 24; collinear features make explanations unstable | Phase 2, should |
| 8 | Models | Keep the forest (regularised: min_samples_leaf 3 to 5, max_features 0.2 to 0.3, capped depth), SVM and logistic regression. Add ExtraTrees, gradient boosting (HistGradientBoosting, XGBoost or LightGBM), a soft-voting ensemble and calibrated probabilities (Platt or isotonic). One reference with frozen pre-trained CNN embeddings and logistic regression, to show what hand-crafted features give up | Forest train accuracy 1.000 against a validation plateau of 0.54 to 0.59 | Phase 2, should |

| # | Stage | What to do | Evidence it answers | Priority |
|---|---|---|---|---|
| 9 | Augmentation | Train only. Geometric copies (flips, 90 degree turns, crops) and photometric copies (brightness, contrast, gamma, shadow, colour cast) of training images; features recomputed for each copy; copies keep the source group. Compare none, geometric, and geometric plus photometric on clean and perturbed originals | Tests whether robustness comes from exposure in training; small dataset | Phase 2, should |
| 10 | Patch-based samples | Resize every image to one scale, cut 4 to 9 overlapping patches (for example 128 px) from the leaf region, extract features per patch, train at patch level and average the probabilities per image. Splits stay at source-photo level | 4 to 9 times more training samples, removes the resolution mismatch, gives spatial heatmaps for explanation | Phase 3, could |
| 11 | Scores | Headline macro-F1; add balanced accuracy, MCC, per-class F1, expected calibration error, Brier score and a selective-prediction curve (accuracy against coverage). Robustness as perturbed divided by clean, area under the severity curve, and worst case. Always mean and std over splits | Single-split numbers are unreliable | Phase 1, should |

**Explainability study (the research contribution).**

| Part | What to do | What it shows |
|---|---|---|
| Importance methods | Impurity, permutation, TreeSHAP (forest, boosting), linear SHAP (logistic regression), kernel SHAP or permutation (SVM), LIME on feature vectors, partial dependence for the top features | Whether methods agree on what matters |
| Correlated features | Hierarchical clusters of features (one minus absolute Spearman correlation); importance per cluster (10 to 15 clusters instead of three groups); cluster-wise permutation | Stability that is not an artefact of collinearity |
| Stability | Spearman, Kendall, rank-biased overlap, top-k Jaccard and the Kuncheva index, across bootstrap resamples, seeds, perturbations, models, masks and resolution variants, each with an interval | How much an explanation changes under each source of variation |
| Faithfulness | Deletion and insertion curves (permute the top k against k random features); label-randomisation sanity check (importance should collapse); agreement between methods | Whether the explanations describe the model |
| Shortcut audit | Correlate each feature and each attribution with image metadata; compare the model with and without shortcut-linked features; check whether explanations point at them | Whether the explanation exposes the shortcut |
| Plausibility | Compare attributions with symptoms (lesion colour, ring pattern, yellowing) and with lesion masks; patch-level heatmaps against lesion regions | Whether the cues are medically meaningful |
| Data size | Repeat the stability measures at 25, 50, 75 and 100% of the training data | Whether more data would stabilise explanations |

**Claims we can defend if the plan works.**

- A leakage-safe protocol for a small field dataset, with each pitfall measured: duplicates across classes, copies across splits, a source shortcut and single-split noise.
- How lighting and masking change the accuracy and the explanations of classical models, with train-time augmentation compared against test-only evaluation.
- How stable explanations are across folds, perturbations, models, masks and resolution, measured at the level of correlated feature clusters.

**Schedule.** Today is Oct 14; the mid-semester report is due Nov 13 and the paper Dec 24.

| Phase | Dates | Work | Owners |
|---|---|---|---|
| Phase 1 | Oct 15 to 24 | Stages 1, 2, 3, 4 and 11 | A: stages 2 to 4 (data, perturbations); B: stage 1; C: stage 11 and the report |
| Phase 2 | Oct 25 to Nov 8 | Stages 5 to 9 | A: stages 5 and 6; B: stages 7 to 9; C: evaluation scripts and the report |
| Mid-semester report | Nov 9 to 13 | Writing and the literature survey | All |
| Phase 3 | Nov 14 to Dec 10 | Stage 10, the explainability study, an external test set, significance tests | C leads the explainability study; A the external data; B the models |
| Paper | Dec 11 to 24 | Writing and checks | All |

## 16. Dataset size, overfitting and external data

The set has 813 images (570 for training), the random forest fits its training data perfectly (accuracy 1.000), and validation F1 stops improving after about 225 training images. More images of the same kind and source would help less than more varied, independently sourced images and fewer shortcuts.

| Option | What it addresses | Risks and checks | Priority |
|---|---|---|---|
| Train-only augmentation | Variance, invariance to lighting | Features recomputed per copy; copies keep their group; never augment validation or test | High |
| Stronger regularisation and feature selection | Overfitting | Selection and tuning inside each training fold only | High |
| Augmented Dataset as extra training copies only | Size of the training set | Map each augmented file to its raw source by file stem and hash; use only siblings of training-split photos; adds no new information and carries the same shortcut | Medium |
| Patch-based samples | 4 to 9 times more samples; resolution mismatch | Patches share their photo's group; average per image | Medium |
| Class weights, or SMOTE on feature vectors | The 2.4x imbalance (80 to 191 images per class) | SMOTE only inside training folds; compare against weights | Medium |
| Frozen pretrained embeddings | Small data | Stronger reference but less transparent; keep as one baseline | Medium |
| Own photos as an external test set | Independence and source control | 100 to 200 leaves per class with a fixed protocol (phone, distance, daylight) and logged metadata; never mixed into training | High if feasible |
| Public tomato datasets as external test sets | Domain shift | Table below; check licences and hash-match first | High, as a test and not for merging |

| Option | What it addresses | Risks and checks | Priority |
|---|---|---|---|
| Semi-supervised or pseudo-labelling | Unlabelled field images | Needs unlabelled images; risks confirming the shortcut | Low |
| Synthetic images (GAN or diffusion) | Size | Adds artefacts and is hard to defend in a paper | Low |

**Candidate external datasets (facts read from the sources below on Oct 14, 2026).**

| Dataset | What the source says | Use here |
|---|---|---|
| PlantDoc | 2,598 images, 27 classes over 13 species; scraped from search engines and filtered by hand; 8 tomato classes including all six of ours; no licence named in the paper | Best class match, but web-sourced like parts of our data: hash-match against our images first |
| Tomato Leaf Dataset (Bangladesh) | 731 images, 7 classes, Canon EOS M50, tomato fields; resized to 640 x 640; shares 4 of our classes (early blight, late blight, leaf mold, healthy); the article is CC BY, the dataset licence is not confirmed | Independent camera and country: a good external test on 4 classes |
| Tomato-Village | Field images from Rajasthan, India, with classification and detection variants; classes, size and licence not confirmed from the paper page | Check that it is available before planning on it |
| PlantVillage | Not checked in this session; commonly described as laboratory-style images | Only as a contrast to show the lab-to-field gap |

Do not merge datasets into one pool before the source controls in section 15 are done: each dataset has its own camera and background, so the class could become predictable from the dataset it came from. Train on ours and test on theirs, or train on several with

leave-one-dataset-out evaluation. Some file names in our raw set look like annotation-tool exports (one ends in .rf. followed by a hash) and others like web downloads, so overlap with PlantDoc is possible and has to be checked.

Sources: PlantDoc paper, Pakistan dataset page, Tomato Leaf Dataset article, Tomato-Village paper.